

6G PGD Class

Transcript

<https://otter.ai/u/dBBHNrNwpB99SDYHCdf-DAVe9kw?view=summary>

Overview

Prof. Neelkanth Kundu discussed quantum teleportation, explaining the process using Alice and Bob's entangled qubits. Ajay Pratap Singh questioned the interpretation of bits in the teleportation protocol, which Prof. Kundu clarified, emphasizing the entanglement between Alice and Bob's qubits. The session also covered quantum capacity, defining it as the maximum amount of quantum information that can be transmitted per channel use. The coherent information, calculated as the difference between Bob's quantum state entropy and the combined state entropy, was highlighted as a key metric. The discussion concluded with an overview of quantum key distribution (QKD) and its potential to provide information-theoretic security.

Action Items

- [] @Neel Kanth Kundu - Share the previously recorded lecture notes taken via the portal with the class.

Outline

Teleportation Protocol Clarification

- Prof. Neelkanth Kundu begins the lecture, mentioning he will wait a few more minutes for stragglers.
- Nakul Agrawal requests Prof. Kundu to share his notes taken with a portal, which he agrees to do.
- Ajay Pratap Singh asks a question about the teleportation protocol, specifically about the interpretation of bits in the context of Alice and Bob.
- Prof. Kundu explains the combined quantum system consisting of Alice's arbitrary qubit and the entangled quantum state between Alice and Bob.
- Ajay Pratap Singh and Prof. Kundu discuss the third bit in the teleportation protocol, clarifying that it belongs to Bob but is entangled with Alice's qubit.

Entanglement and Teleportation

- Ajay Pratap Singh raises concerns about the stability of entanglement during the teleportation process.
- Prof. Kundu explains that entanglement is consumed at the end of teleportation, and the qubit at Bob's end collapses to one of four states.
- Ajay Pratap Singh questions whether the state of Bob's qubit changes instantly due to entanglement, even if Alice does not send the information.

- Prof. Kundu confirms that the state of Bob's qubit changes instantly due to entanglement, but it is not faster than light communication.
- The discussion touches on the theoretical implications of entanglement and its counterintuitive nature.

Quantum Teleportation Summary and Noisy Channels

- Prof. Kundu summarizes the quantum teleportation protocol, emphasizing the use of pre-shared entanglement and classical communication channels.
- He explains that in practical scenarios, the entanglement shared between Alice and Bob might be noisy due to noisy channels.
- The concept of fidelity is introduced, which quantifies how well the teleported qubit matches the original state.
- Prof. Kundu mentions that researchers report the fidelity of the teleported qubit to assess the success of the protocol.
- The discussion transitions to the topic of quantum capacity, which involves transmitting quantum information using quantum channels.

Quantum Capacity and Coherent Information

- Prof. Kundu introduces the concept of quantum capacity, which is the maximum amount of quantum information that can be transmitted per channel use.
- He explains that the quantum capacity is lower bounded by the maximum coherent information between Alice and Bob.
- The coherent information is defined as the difference between the von Neumann entropy of Bob's quantum state and the von Neumann entropy of the combined state of Alice and Bob.
- Prof. Kundu provides a mathematical result from quantum information theory research, which is the quantum capacity theorem.
- Nakul Agrawal asks a question about the information leaked to the environment and the eavesdropper, which Prof. Kundu clarifies.

Quantum Capacity of Identity Channel

- Prof. Kundu explains the quantum capacity of the identity channel, which maps the input density operator to the output density operator.
- He states that the quantum capacity of the identity channel is one qubit per channel use.
- The discussion includes the comparison of classical capacity, private capacity, and quantum capacity for noiseless channels.
- Prof. Kundu mentions that the classical capacity can be doubled with entanglement, but quantum capacity remains the same.
- The session concludes with a brief overview of the next module on quantum key distribution, which will cover various QKD protocols and their applications.